BASE CHANGE FOR DERIVED PUSHFORWARD AGAINST A GIVEN MAP TO K'

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source proves the scheme case by the corresponding Derived Categories of Schemes lemma and carries out the induction step in full with an explicit commutative diagram, closing with the third-isomorphism lemma for distinguished triangles

Editorial note. The label lemma-single-complex-base-change names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- cartesian diagrams of algebraic spaces and the base change morphism Lg^*Rf_*K to Rf'_*L(g')^*K
- the base change condition of the condition lemma (counted earlier in this lane) for a single complex K with a map L(g')^*K to K'
- the induction principle for properties of algebraic spaces etale locally
- relative Mayer-Vietoris triangles for distinguished squares

Source reference. Original lemma 014A: sources/stacks-chow/tags.txt, source lines 1465–1465. Prerequisite pointer invoked at proof line(s) 5562; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at proof line(s) 5525; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08DR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1563–1563. Prerequisite pointer invoked at proof line(s) 5514; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DJ8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5365–5365. Prerequisite pointer invoked at proof line(s) 5503; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 08HZ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1979–1979. Prerequisite pointer invoked at proof line(s) 5548; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DJ9: sources/stacks-chow/tags.txt, source lines 16969–16969. Prerequisite pointer invoked at proof line(s) 5536; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DJ8: sources/stacks-chow/tags.txt, source lines 16968–16968. Prerequisite pointer invoked at proof line(s) 5528; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08GH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 896–896. Prerequisite pointer invoked at proof line(s) 5532; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian diagram of algebraic spaces over S. Let $K \in D_{QCoh}(\mathcal{O}_X)$ and let $L(g')^*K \to K'$ be a map in $D_{QCoh}(\mathcal{O}_{X'})$. If

(1) *the equivalent conditions of Lemma 0DJ8 hold, and*
(2) *f is quasi-compact and quasi-separated,*

then the composition $Lg^*Rf_*K \to Rf'_*L(g')^*K \to Rf'_*K'$ is an isomorphism.

Proof. To check the map is an isomorphism we may work étale locally on Y'. Hence we may assume $g : Y' \to Y$ is a morphism of affine schemes. In this case, we will use the induction principle of Lemma 08DR to prove that for a quasi-compact and quasi-separated algebraic space U étale over X the similarly constructed map $Lg^*R(U \to Y)_*K|_U \to R(U' \to Y')_*K'|_{U'}$ is an isomorphism. Here $U' = X' \times_{g',X} U = Y' \times_{g,Y} U$.

If U is a scheme (for example affine), then the result holds. Namely, then Y, Y', U, U' are schemes, K and K' come from objects of the derived category of the underlying schemes by Lemma 071Q and the condition of Derived Categories of Schemes, Lemma 0DJ8 holds for these complexes by Lemma 0DJ8. Thus (by the compatibilities explained in Remark 08GH) we can apply the result in the case of schemes which is Derived Categories of Schemes, Lemma 0DJ9.

The induction step. Let $(U \subset W, V \to W)$ be an elementary distinguished square with W a quasi-compact and quasi-separated algebraic space étale over X, with U quasi-compact, V affine and the result holds for U, V, and $U \times_W V$. To easy notation we replace W by X (this is permissible at this point). Denote $a : U \to Y$, $b : V \to Y$, and $c : U \times_X V \to Y$ the obvious morphisms. Let $a' : U' \to Y'$, $b' : V' \to Y'$ and $c' : U' \times_{X'} V' \to Y'$ be the base changes of a, b, and c. Using

the distinguished triangles from relative Mayer-Vietoris (Lemma 08HZ) we obtain a commutative diagram

$$\begin{array}{ccc}
Lg^*Rf_*K & \longrightarrow & Rf'_*K' \\
\downarrow & & \downarrow \\
Lg^*Ra_*K|_U \oplus Lg^*Rb_*K|_V & \longrightarrow & Ra'_*K'|_{U'} \oplus Rb'_*K'|_{V'} \\
\downarrow & & \downarrow \\
Lg^*Rc_*K|_{U\times_X V} & \longrightarrow & Rc'_*K'|_{U'\times_{X'} V'} \\
\downarrow & & \downarrow \\
Lg^*Rf_*K[1] & \longrightarrow & Rf'_*K'[1]
\end{array}$$

Since the 2nd and 3rd horizontal arrows are isomorphisms so is the first (Derived Categories, Lemma 014A) and the proof of the lemma is finished. □